

Executive Summary — pqc_crypto_module v0.1.0

| **Disclaimer:** Prepared for FIPS 140-3 evaluation, not currently validated.

What We Are

pqc_crypto_module is a standalone Rust-based post-quantum cryptographic module developed for the **Cerulean Ledger** distributed ledger platform. It isolates all cryptographic operations behind a strict API boundary with approved-mode enforcement, startup self-tests, and automatic key zeroization.

What It Implements

Algorithm	Standard	Purpose
ML-DSA-65	FIPS 204	Digital signatures (key generation, signing, verification)
ML-KEM-768	FIPS 203	Key encapsulation and decapsulation
SHA3-256	FIPS 202	Cryptographic hashing

Non-approved legacy algorithms (Ed25519, SHA-256, HMAC-SHA256) are present for backward compatibility with pre-PQC ledger data. They are runtime-gated and blocked when the module operates in approved mode. A compile-time feature flag (`approved-only`) removes them entirely.

Why It Matters

The transition to post-quantum cryptography is a pressing requirement for organizations handling sensitive data with long-term confidentiality needs. NIST has finalized FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA) as the first post-quantum standards. Cerulean Ledger is built from the ground up with PQC as the default, positioning it ahead of the migration curve mandated by NSM-10 and OMB M-23-02.

Current Readiness

- **Architecture:** Clean cryptographic boundary (11 source files, single public API module).
- **Documentation:** 9 FIPS 140-3 documentation artifacts produced (Security Policy, Design Document, FSM, Key Management, Self-Tests, Non-Approved Usage, Boundary Definition, Reproducible Build, Operational Guidance).
- **Testing:** 1500+ tests across 12 test suites; 100% boundary compliance verified.
- **Self-Tests:** Known Answer Tests (KATs) for all approved algorithms run at module initialization; module enters error state on failure.
- **Key Management:** All key types implement ZeroizeOnDrop; keys are never exposed outside the boundary.

What We Are Seeking

We are seeking engagement with an NVLAP-accredited laboratory for **FIPS 140-3 Level 1 validation** of `pqc_crypto_module`. Specifically:

1. **CAVP algorithm certificate testing** for ML-DSA-65, ML-KEM-768, and SHA3-256.
2. **FIPS 140-3 module validation** at Security Level 1 (software-only).
3. **Guidance** on NIST ACVP test vector availability for post-quantum algorithms.
4. **Consultation** on SP 800-90B entropy source compliance for `OsRng`.

Contact

[Organization contact details to be filled in before sending]

For technical details, see the accompanying `module_overview.md`.